Introduction to Machine Learning

Week 10

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1. (1 Mark) In a clustering evaluation, a cluster C contains 50 data points. Of these, 30 belong to class A, 15 to class B, and 5 to class C. What is the purity of this cluster?

   (a) 0.5

   (b) 0.6

   (c) 0.7

   (d) 0.8

   Soln. B - Purity = (Number of data points in the most frequent class) / (Total number of data points)

2. (1 Mark) Consider the following 2D dataset with 10 points:

   $$(1,1),(1,2),(2,1),(2,2),(3,3),$$
   $$(8,8),(8,9),(9,8),(9,9),(10,10)$$

   Using DBSCAN with $\epsilon = 1.5$ and MinPts = 3, how many core points are there in this dataset?

   (a) 4

   (b) 5

   (c) 8

   (d) 10

   Soln. C - To be a core point, it needs at least 3 points (including itself) within $\epsilon = 1.5$ distance. There are 8 core points: (1,1), (1,2), (2,1), (2,2) from first group and (8,8), (8,9), (9,8), (9,9) from second group.

3. (1 Mark) In BIRCH, using number of points **N**, sum of points **SUM** and sum of squared points **SS**, we can determine the centroid and radius of the combination of any two clusters A and B. How do you determine the radius of the combined cluster? (In terms of **N**,**SUM** and **SS** of both two clusters A and B)

   Radius of a cluster is given by:

   $$Radius = \sqrt{\frac{SS}{N} - (\frac{SUM}{N})^2}$$

   Note: We use the following definition of radius from the BIRCH paper: *"Radius is the average distance from the member points to the centroid."*

   (a) $Radius = \sqrt{\frac{SS_A}{N_A} - (\frac{SUM_A}{N_A})^2 + \frac{SS_B}{N_B} - (\frac{SUM_B}{N_B})^2}$

   (b) $Radius = \sqrt{\frac{SS_A}{N_A} - (\frac{SUM_A}{N_A})^2} + \sqrt{\frac{SS_B}{N_B} - (\frac{SUM_B}{N_B})^2}$

(c) $Radius = \sqrt{\frac{SS_A+SS_B}{N_A+N_B} - (\frac{SUM_A+SUM_B}{N_A+N_B})^2}$

(d) $Radius = \sqrt{\frac{SS_A}{N_A} + \frac{SS_B}{N_B} - (\frac{SUM_A+SUM_B}{N_A+N_B})^2}$

Soln. C
$CF_{A+B} = CF_A + CF_B$

Therefore,
$N_{A+B} = N_A + N_B$
$SUM_{A+B} = SUM_A + SUM_B$
$SS_{A+B} = SS_A + SS_B$

Replace the above in the formula of radius given in the question to get the formula for the combined cluster.

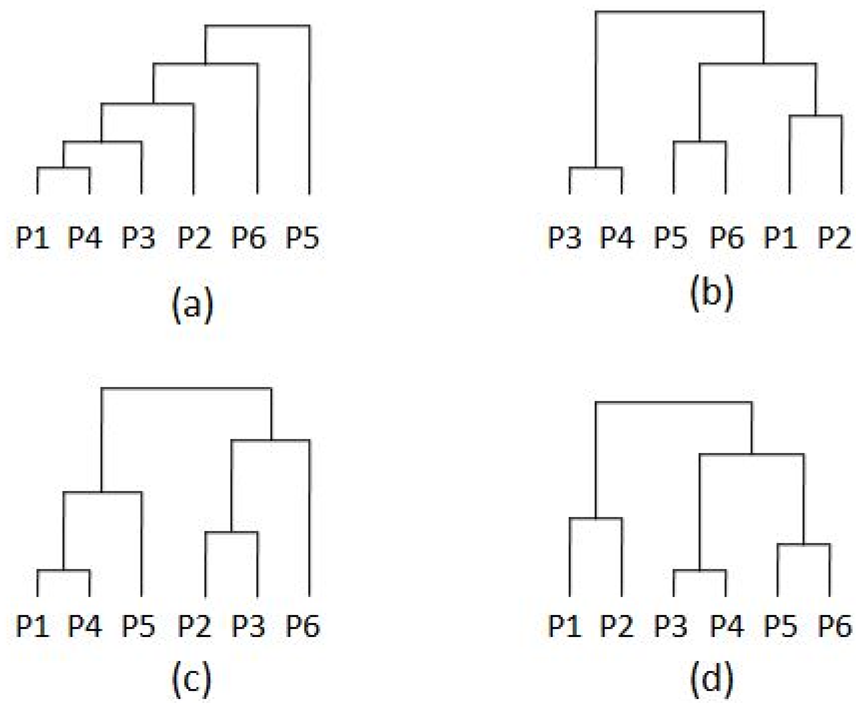
4. (1 Mark) Which of the following properties are TRUE?

   (a) Using the CURE algorithm can lead to non-convex clusters.
   (b) K-means scales better than CURE for large datasets.
   (c) CURE is a simplification of K-means and hence scales better than k-means for large datasets.
   (d) K-means being more expensive to run on large datasets, can give non-convex clusters too.

   Soln. A- Refer to the lecture videos

5. (1 Mark) The pairwise distance between 6 points is given below. Which of the option shows the hierarchy of clusters created by single link clustering algorithm?

	P1	P2	P3	P4	P5	P6
P1	0	3	8	9	5	4
P2	3	0	9	8	10	9
P3	8	9	0	1	6	7
P4	9	8	1	0	7	8
P5	5	10	6	7	0	2
P6	4	9	7	8	2	0

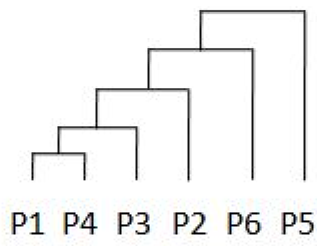


Soln. B
Step 1: Connect closest pair of points. Closest pairs are:
d(P3, P4) = 1
d(P5, P6) = 2
d(P1, P2) = 3

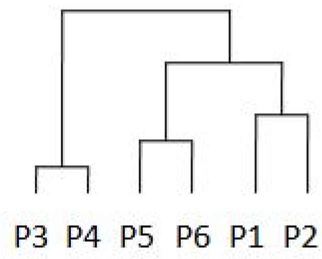
Step 2: Connect clusters with single link. The cluster pair to combine is bolded:
d({P1, P2}, {P3, P4}) = 8
d({P1, P2}, {P5, P6}) = 4
d({P5, P6}, {P3, P4}) = 6

Step 3: Connect the final 2 clusters

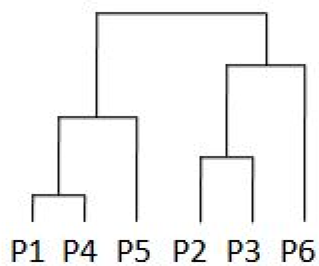
6. (1 Mark) For the pairwise distance matrix given in the previous question, which of the following shows the hierarchy of clusters created by the complete link clustering algorithm.



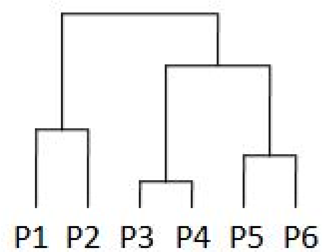
(a)



(b)



(c)



(d)

Soln. B

Step 1: Same as previous question

Step 2: Connect clusters with complete link. The cluster pair to combine is bolded:
d({P1, P2}, {P3, P4}) = 9
d({P1, P2}, {P5, P6}) = 10
d({P5, P6}, {P3, P4}) = 8

Step 3: Connect the final 2 clusters